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C0902810

Python Programming

Task #1

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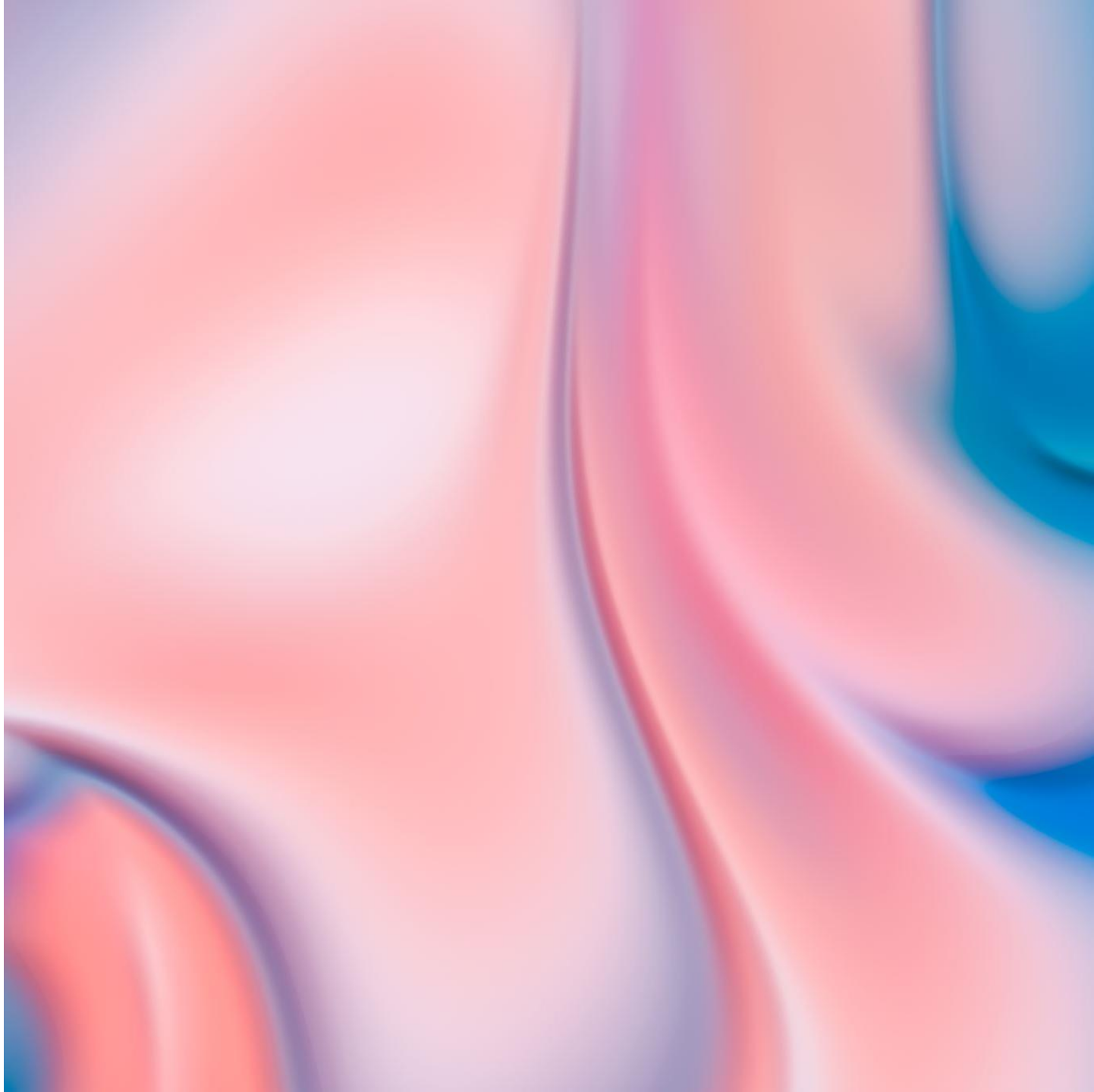

Q.1. Write 2 real-life examples of each feature of Object-Oriented Programming:

- **Abstraction**
- **Class**
- **Object**
- **Encapsulation**
- **Inheritance**
- **Polymorphism**

+ **Abstraction:** Approach of hiding the detailed information of method or class and only showing the essential information.
+ **Class:** Collection of data members and member functions which are accessible using objects created for that class.
+ **Object:** A memory block and an approach for accessing various data members and member functions of the class of which the object has been declared.
+ **Encapsulation:** Bundling the data members and member functions in a class and accessible only using the objects created for that class.
+ **Inheritance:** Process of relating between two or more classes which allows reusability of code.
+ **Polymorphism:** Same name, many forms; an approach using which we can use same name of the method or other data in a class which has different implementation as per the class.

Abstraction – Real Life Example

- Using microwave oven for baking a pizza without knowing how the machine works.
- Using MatPlotLib package in Python and drawing various charts for analysis without knowing how it is built as a library.

Class

- Making a laptop class that has CPU, RAM, Storage, and other useful components.
- Making a class of cars which includes Model, Engine Details, Wheel Info, Storage Capacity, and Color Shade.

Object

- Google-Pixel-7 can be an object of the class Mobile which has its own CPU, Memory, Storage, and Battery.
- Hundai Alentra is a car which has its own car-engine, wheel-info, storage-capacity, and the color shades.

Encapsulation

- A laptop bag which contains various electronic components such as mouse, keyboard, laptop, external disk, and pen-drive.
- A laptop having various internal components such as CPU, RAM, Storage, Webcam, Ports, Screen Detail, Graphics Card (GPU), Trackpad, Heatsink, and Software Support.

Inheritance

- A laptop’s components can be a mouse, keyboard, screen, processing components, and storage; while processing components can contain CPU, GPU, and custom chips.
- Python program is a subset of high-level programming language and so is Java, C, C++, C#, PHP, and JavaScript; while Python has a support to libraries such as Tensor-Flow, PyTorch, and MatPlotLib.

Polymorphism

- Hydrogen can be in solid, liquid, and gases forms applying different chemistry.
- Same snake can behave differently when in a calm environment and in an aggressive angry-making environment.

Q.2. Write answers to all questions from slides of Chapter 1.

- **List 1:**
  - **Q1.** Name a few high-level programming language.
  - **Q2.** Why are there so many programming languages?
  - **Q3.** Which programming languages have survived? Why?
- **List 2:**
  - **Q1.** Let us spend a few minutes to find a few procedural programming languages.
  - **Q2.** Let us spend a few minutes to find a few object-oriented programming languages.
  - **Q3.** Advanced Topic (optional): Try to understand what encapsulation means in OOP and how it can be used in divide and conquer approach for solving various problems.

Q.2.1.1. Name a few high-level programming language.

- Python
- Java
- C
- C++
- C#
- PHP
- R
- JavaScript
- HTML & CSS

Q.2.1.2. Why are there so many programming languages?

Reasons:

- Compatibility issues with the hardware devices and other resources.
- Compilation and interpretation process time.
- User-interface.
- Performance and latency.
- Licensing problems.
- Community support.
- Syntax as a barrier (Ease of performing a task with given programming language).
- Multi-tasking and asynchronous programming support.
- Architecture and programming paradigms support (OOP, POP, etc).

Q.2.1.3. Which programming languages have survived? Why?

- Java and C are those languages that has survived due of the fact that when these were in-demand, there were a lot of software and applications developed and now to maintain them, it is necessary to have developers to work on it.

Q.2.2.1. Let us spend a few minutes to find a few procedural programming languages.

- C, C++, Java, Pascal, BASIC, ALGOL, and COBEL are some of the procedural programming languages that uses step-by-step approach in order to execute the business logic implemented by a programmer.

Q.2.2.2. Let us spend a few minutes to find a few object-oriented programming languages.

- Python, Java, C++, JavaScript, PHP, C#, and R are some of the object-oriented programming languages that uses objects and classes while allowing the use of other OOP fundamentals and features in order to execute the business logic implemented by a developer.

Q.2.2.3. Advanced Topic (optional): Try to understand what encapsulation means in OOP and how it can be used in divide and conquer approach for solving various problems.

- Encapsulation is a process of wrapping up the data member and member methods in a class which are accessible using objects created for that class.
- Divide and conquer is an approach on how a problem can be divided into sub-problems till it can not be divided further, solving those sub-problems, and gradually combining them in order to solve the actual problem.
- This approach not only makes it easier to understand the problem and divide it in as many resources as we are able to, but also can be completed faster and more effectively when the given problem is too big to solve and allows parallelism.
- Because encapsulation promotes the use of class which has various data members and member functions, it becomes easier to deploy a separate method for each individual small problems a class can consist of with provided arguments for variations.

Q.2.2.3. Example:

- When searched a question on a search engine (such as Google) provides a list of various answers but does not provide the information regarding from where did it searched or how did it gather and filtered-out the data for the user, simply shows the results.
- There would be various sub-processes happening parallelly in the background which would be dividing the query into various possible meaningful and then finding each individual results accordingly while conquering at the end with all of the results found.
- Everything happening over here is encapsulated and hidden from the user, and due to that fact it is not possible to predict whether one instance (object) performed this task or several together collectively for the same problem.

Q.2.2.3. Another Example:

- There is a system which performs image-processing on a given image to find all the objects present in it.
- In this process, the system divides the given image in various segments in order to find various objects present in it and this can be a big hierarchy when the system is doing on an image that has too many objects.
- However, the process is hidden and with the use of various methods present in a class can be helpful in dividing the image in more parts until it finds an object and then combines their results collectively showcasing a list of objects detected in that image.